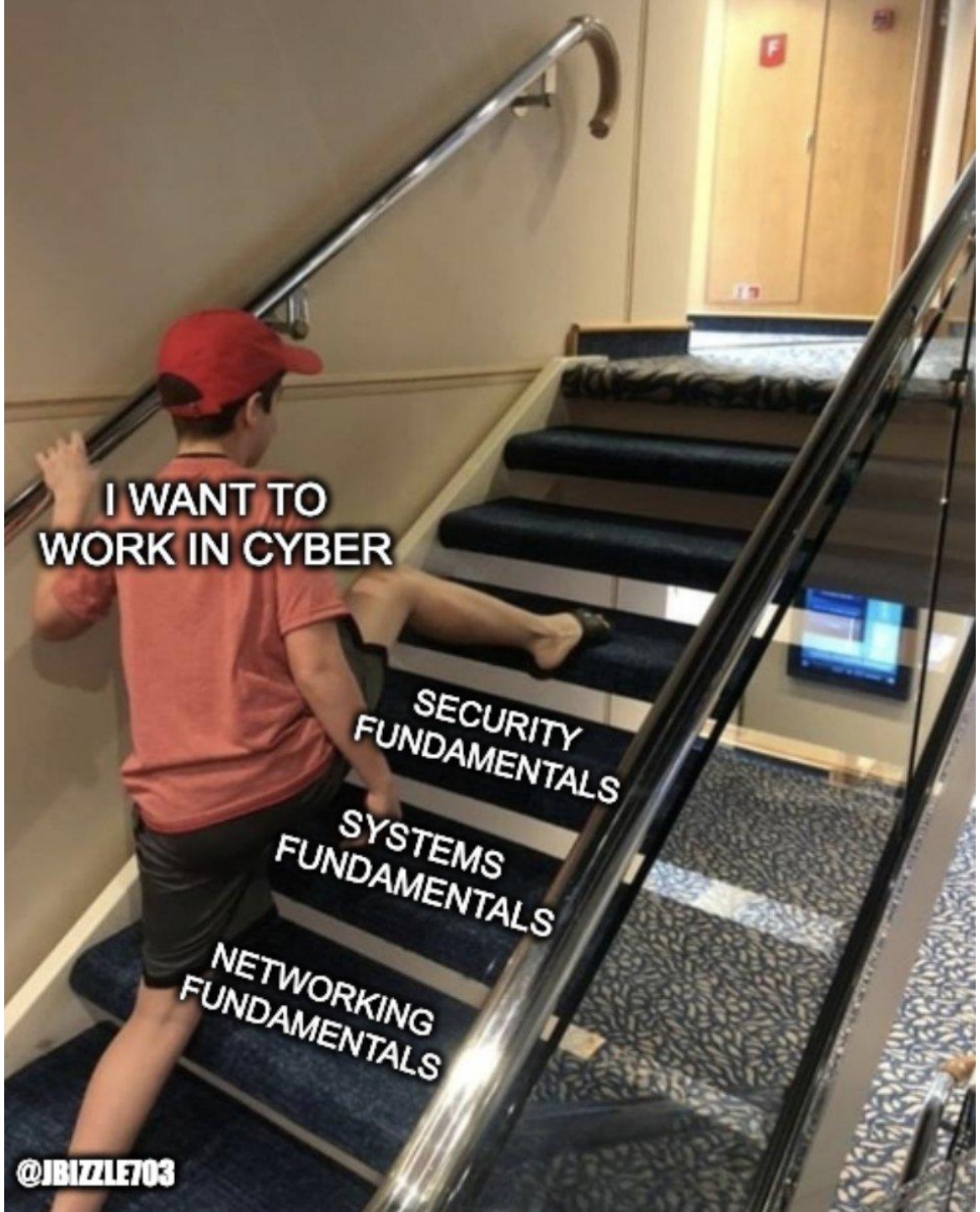
Systemadministration

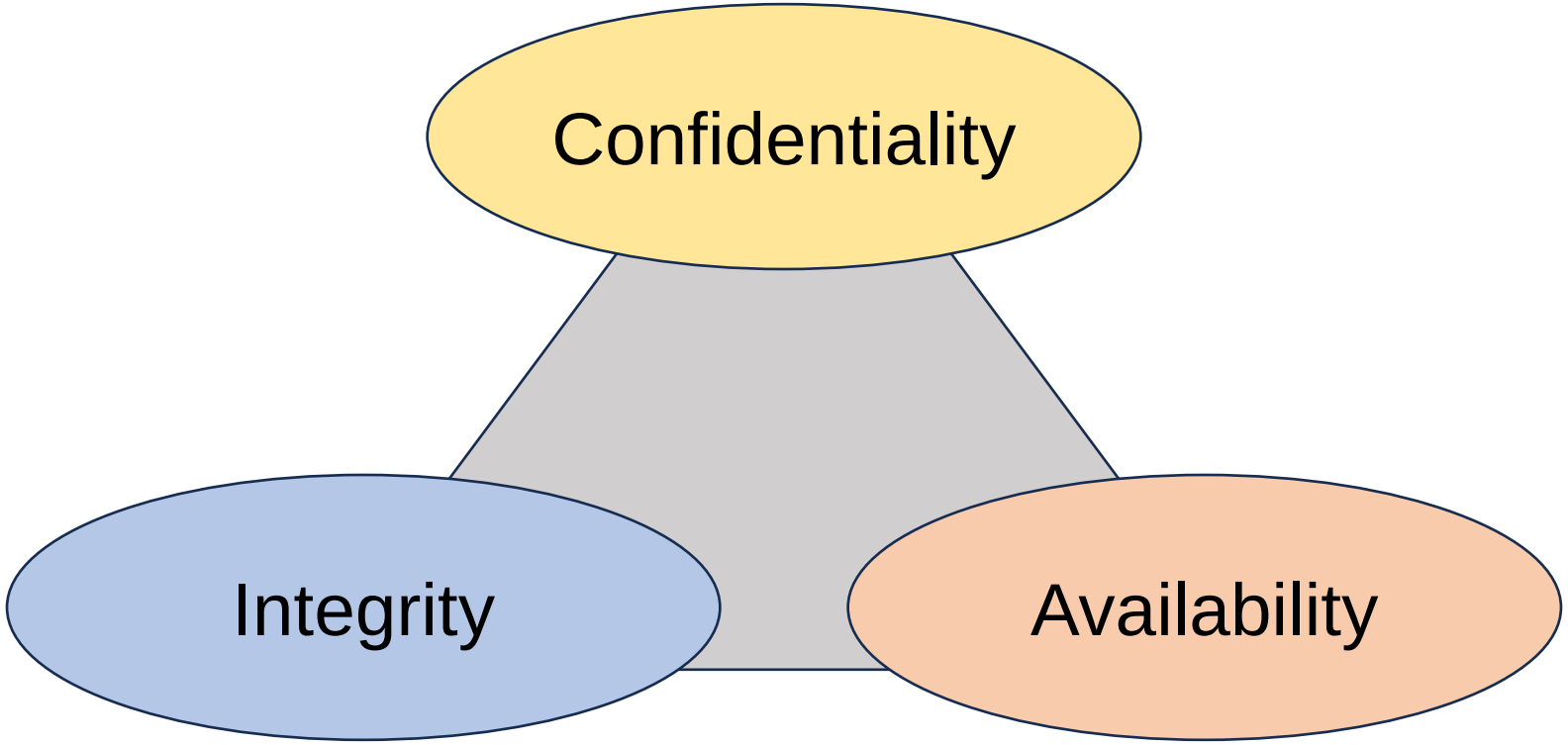
Introduktion

Cybersäkerhet



https://mcusercontent.com/5e5640dc2e2a939f35bf54df2/images/6f3564dc-d02e-83c4-04d4-5ab9bc422f63.jpg

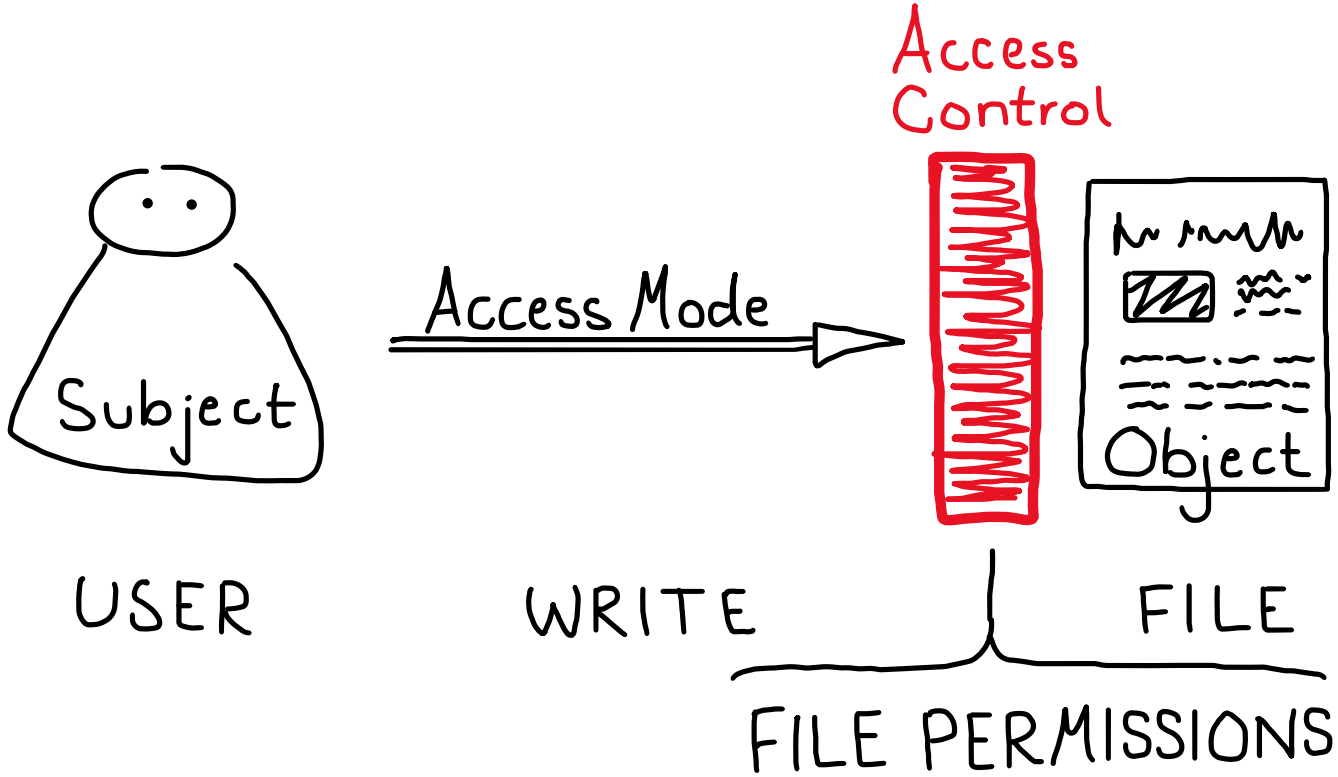
CIA-triad



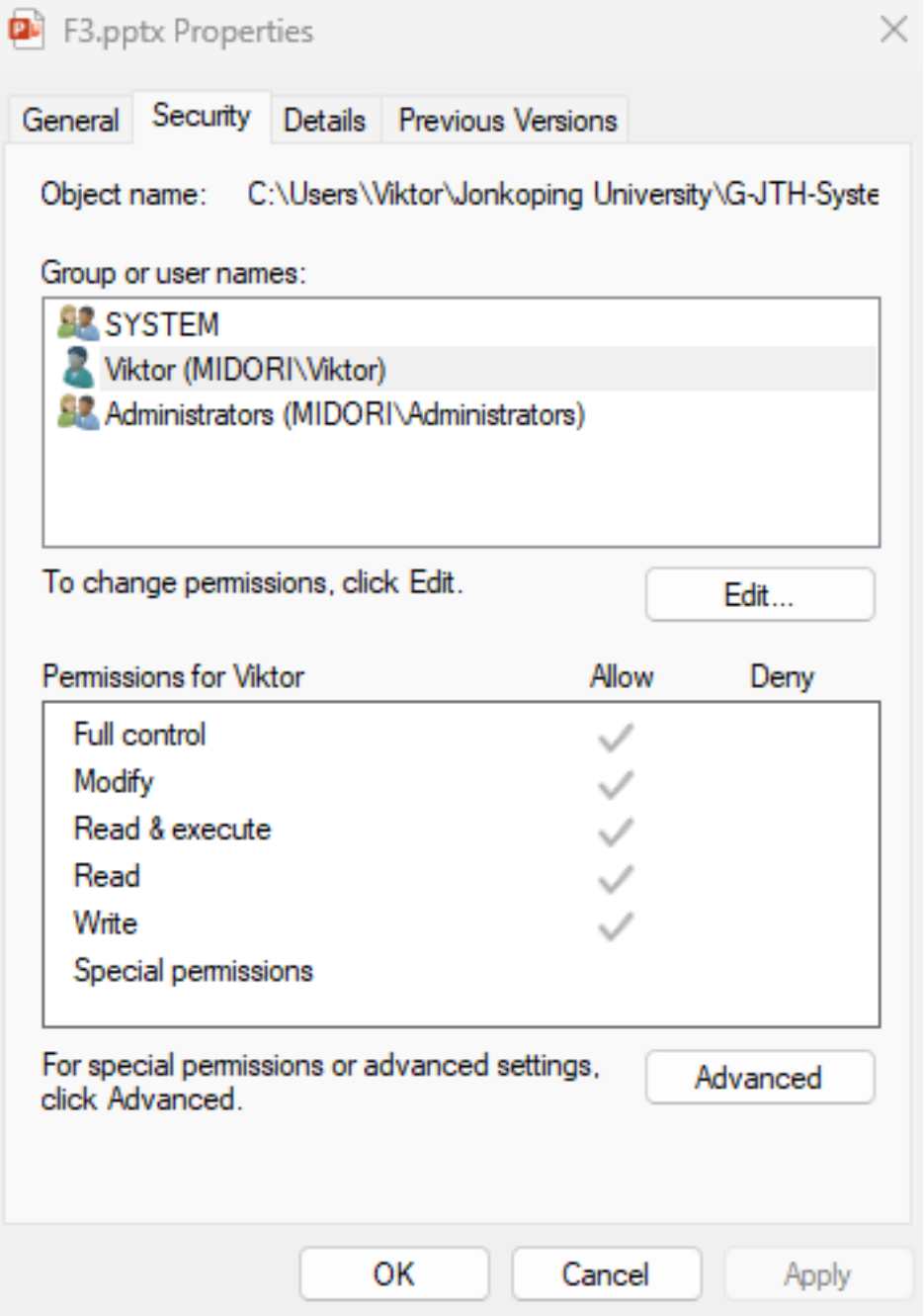
Access Control

Access Control

Ex.



File Permissions



File Permissions

Vad händer om ett nytt OS startas på samma dator?

Live Operating Systems

Behöver inte installeras

Bootar från USB-minne, CD/DVD-skiva

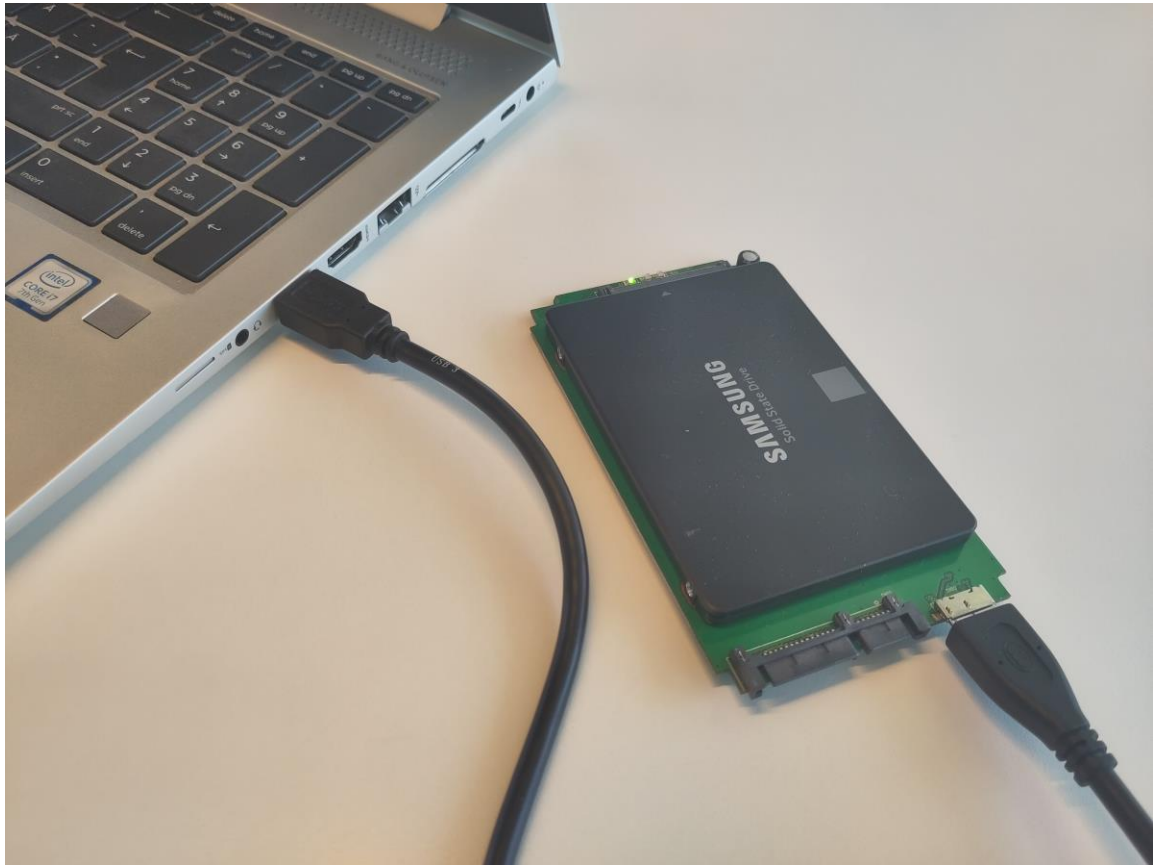
Körs enbart i RAM

This is the logo of Clonezilla, by CloneZilla, Wikimedia Commons (https://commons.wikimedia.org/wiki/File:CZLogo2.png). GFDL 1.2

Disk Encryption

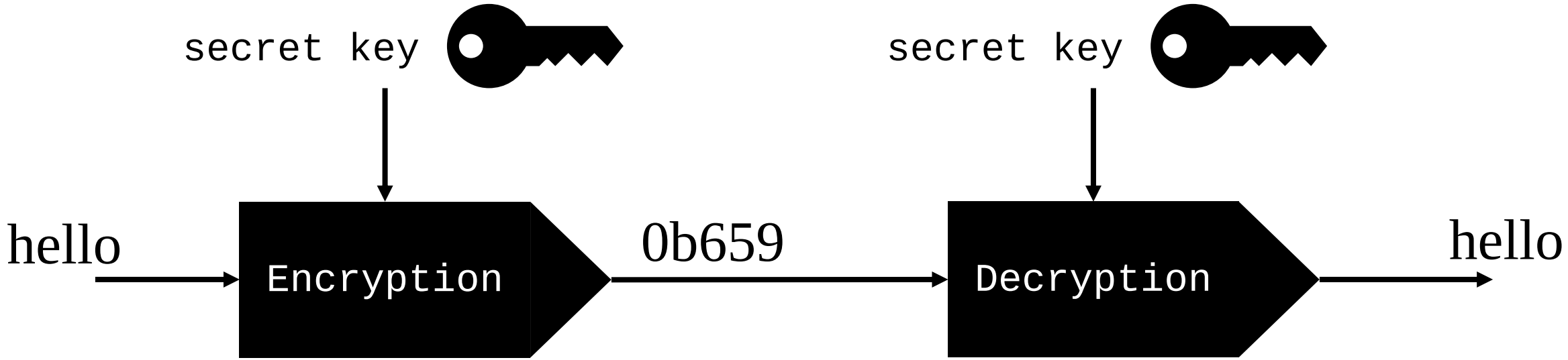
File Permissions

Vad händer om lagringen flyttas till en annan dator?



File Encryption

Förenklat krypterings exempel:



Full Disk Encryption

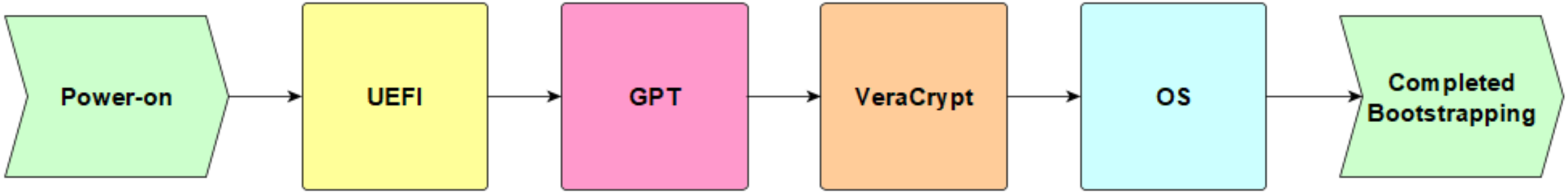
Screenshot of BitLocker icon, by Lisa, Wikimedia Commons (https://en.wikipedia.org/wiki/File:BitLocker_icon.png). Fair use/I enlighet med citaträtt, trademark of Microsoft Corporation.

Microsoft BitLocker

Logo von VeraCrypt, by IDRIX, 2016, Wikimedia Commons (https://commons.wikimedia.org/wiki/File:VeraCrypt_Logo.svg). Public domain, trademark of IDRIX.

VeraCrypt

A veracrypt boot loader tested using VMware Workstation Pro of Windows 10, by MartinHero13, 2021, Wikimedia Commons (https://commons.wikimedia.org/wiki/File:Veracrypt_Boot_Loader.png). CC BY-SA 4.0 International.



Self-Encrypting Drives

Exos X20

Performance, Sustained Data Rate (SDR)

Up to 285 MB/s

Technology

Helium

SED-FIPS Options

Seagate Secure™ SED, Seagate Secure SED- FIPS, Common Criteria

MTBF

2.5M hours

Limited Warranty

5 years

https://www.seagate.com/gb/en/products/enterprise-drives/exos-x/

threat post

Samsung Shattered Encryption on 100M Phones

Author:
Lisa Vaas

February 23, 2022
/ 4:29 pm

One cryptography expert said that 'serious flaws' in the way Samsung phones encrypt sensitive material, as revealed by academics, are 'embarrassingly bad.'

https://threatpost.com/samsung-shattered-encryption-on-100m-phones/178606/

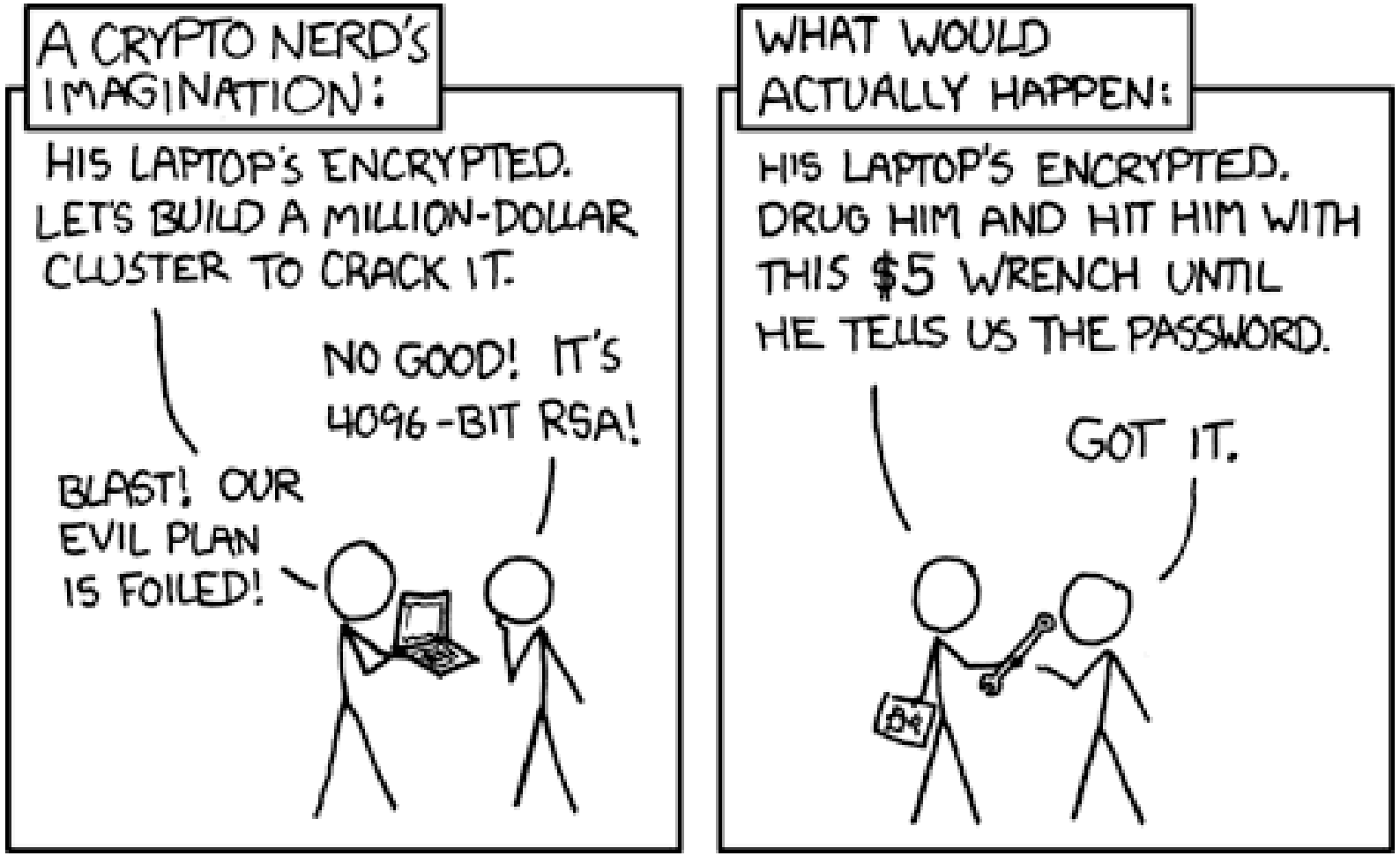
ZD NET

Flaws in self-encrypting SSDs let attackers bypass disk encryption

Master passwords and faulty standards implementations allow attackers access to encrypted data without needing to know the user-chosen password.

Written by **Catalin Cimpanu,** Contributor on Nov. 5, 2018

https://www.zdnet.com/article/flaws-in-self-encrypting-ssds-let-attackers-bypass-disk-encryption/



https://xkcd.com/538/

Malware

Malicious Soft**ware**

Skadligkod, ’virus’

I form av/infekterar existerande:

- Program
- Drivrutiner
- OS-moduler, -kärnan
- Firmware

ars TECHNICA

Trojanized versions of PuTTY utility being used to spread backdoor

Threat actor has connections to hackers backed by North Korean government.

DAN GOODIN - 9/16/2022, 2:37 AM

Getty Images

Enlarge

Researchers believe hackers with connections to the North Korean government have been pushing a Trojanized version of the PuTTY networking utility in an attempt to backdoor the network of organizations they want to spy on.

https://arstechnica.com/information-technology/2022/09/trojanized-versions-of-putty-utility-being-used-to-spread-backdoor/

WIRED

ANDY GREENBERG SECURITY SEP 19, 2023 10:00 AM

Chinese Spies Infected Dozens of Networks With Thumb Drive Malware

Security researchers found USB-based Sogu espionage malware spreading within African operations of European and US firms.

PHOTOGRAPH: JEFFREY HAMILTON/GETTY IMAGES

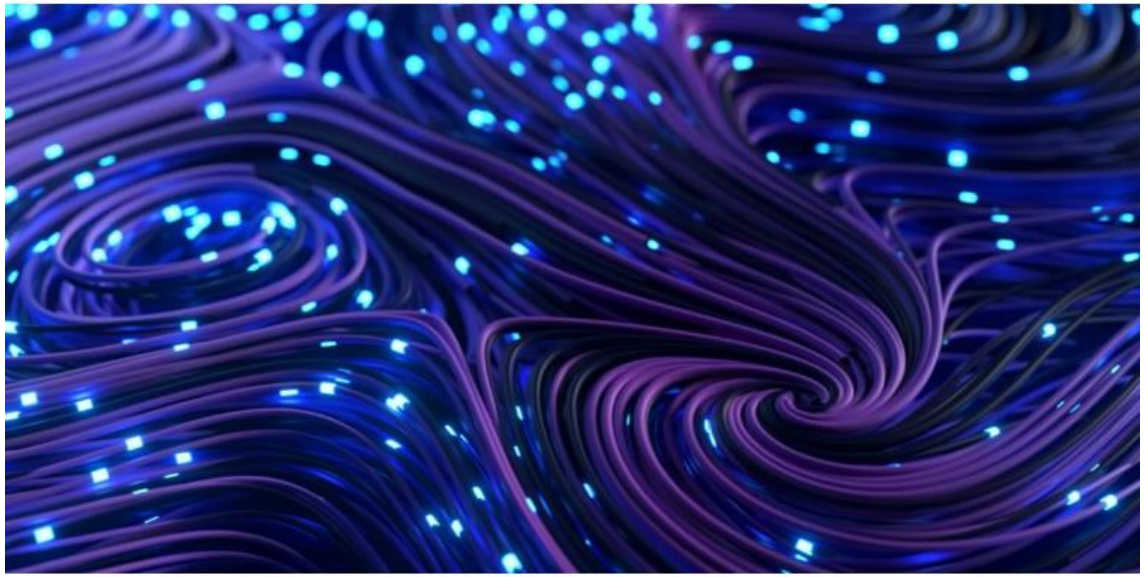
https://www.wired.com/story/china-usb-sogu-malware/

https://www.welivesecurity.com/2018/09/27/lojax-first-uefi-rootkit-found-wild-courtesy-sednit-group/

SECURELIST by Kaspersky

CosmicStrand: the discovery of a sophisticated UEFI firmware rootkit

APT REPORTS 25 JUL 2022



// AUTHORS

Expert GREAT

https://securelist.com/cosmicstrand-uefi-firmware-rootkit/106973/

SECURELIST by Kaspersky

MoonBounce: the dark side of UEFI firmware

APT REPORTS 20 JAN 2022

// AUTHORS

MARK LECHTIK Expert VASILY BERDNIKOV DENIS LEGEZO Expert ILYA BORISOV

https://securelist.com/moonbounce-the-dark-side-of-uefi-firmware/105468/

Menu THE CONVERSATION Sign in

Picking off hard drive manufacturers, one by one. Kaspersky Lab

Malware infecting hard disk firmware remained hidden for 15 years – but who's responsible?

Published: February 19, 2015 3.36pm GMT

Alan Woodward, *University of Surrey*

https://theconversation.com/malware-infecting-hard-disk-firmware-remained-hidden-for-15-years-but-whos-responsible-37774

BLEEPINGCOMPUTER

Firmware attack can drop persistent malware in hidden SSD area

By Bill Toulas

December 30, 2021 09:55 AM

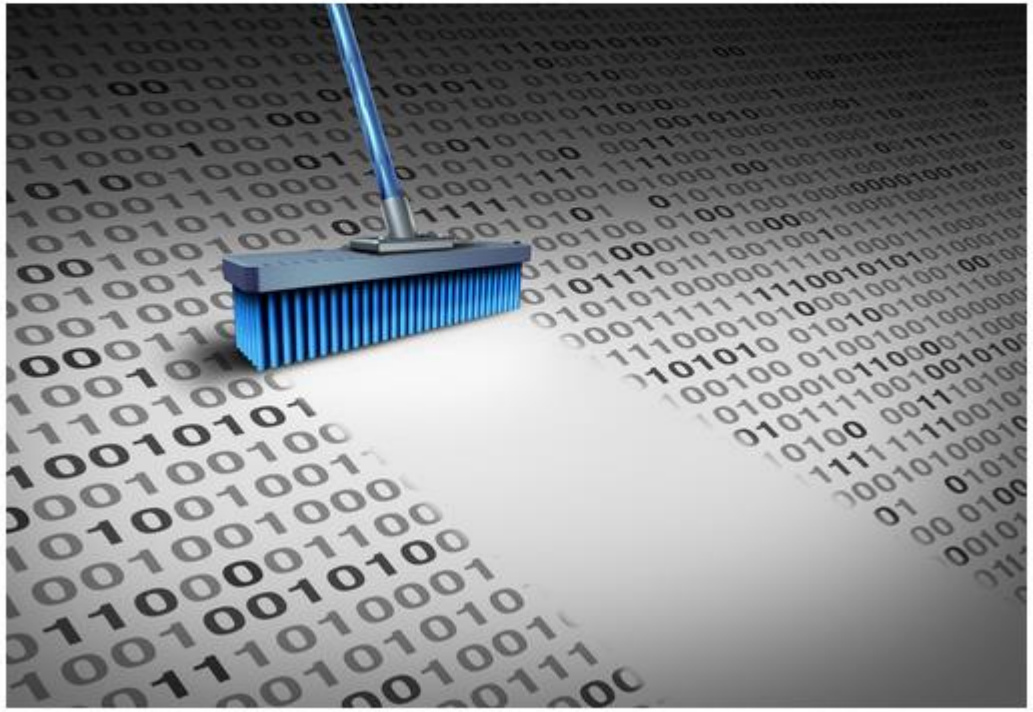
Korean researchers have developed a set of attacks against some solid-state drives (SSDs) that could allow planting malware in a location that's beyond the reach of the user and security solutions.

https://www.bleepingcomputer.com/news/security/firmware-attack-can-drop-persistent-malware-in-hidden-ssd-area/

CYWARE SOCIAL

iLOBleed Rootkit Hides Inside HP Devices to Wipe Off Iranian Data

- January 13, 2022



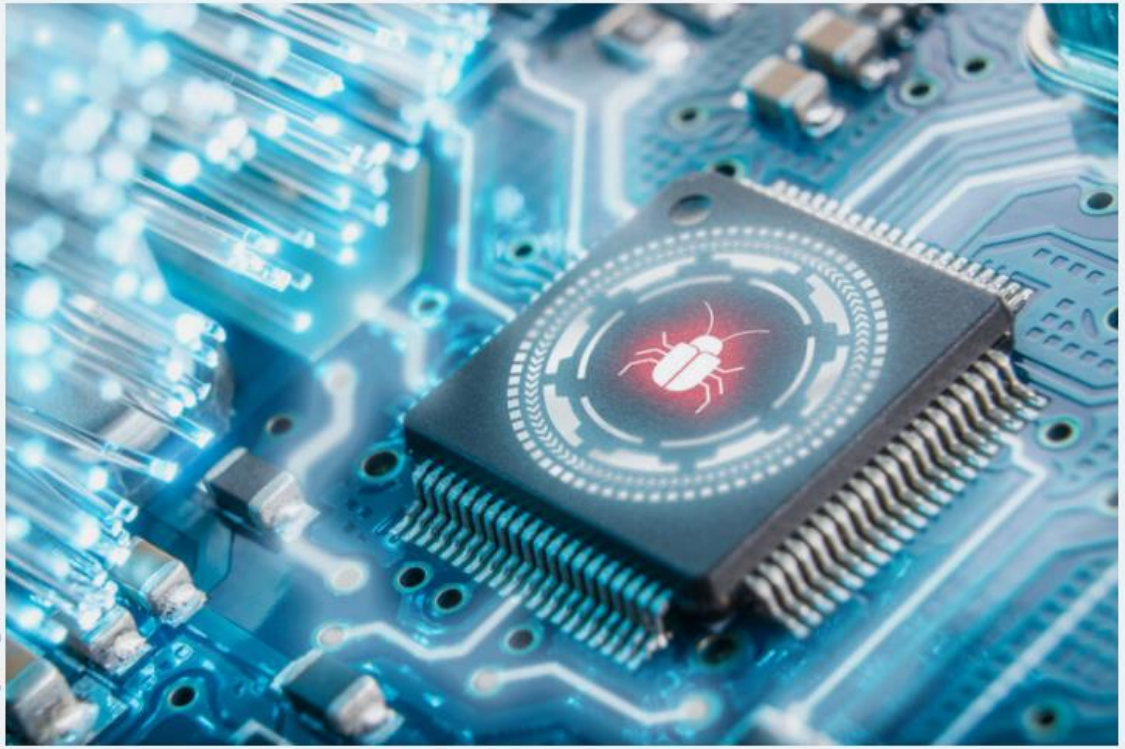
https://cyware.com/news/ilobleed-rootkit-hides-inside-hp-devices-to-wipe-off-iranian-data-0dbbcd81

ars TECHNICA

Vulnerabilities in Supermicro BMCs could allow for unkillable server rootkits

With the ability to manage huge fleets of servers, BMCs are ideal places to stash malware.

DAN GOODIN - 10/5/2023, 12:21 AM



Getty Images

https://arstechnica.com/security/2023/10/vulnerabilities-in-supermicro-bmcs-could-allow-for-unkillable-server-rootkits/

UEFI Secure Boot

UEFI Secure Boot

Enbart kryptografiskt betrodda OS kan startas

Digitalt signerad

- Boot loader
- Kernel

UEFI Secure Boot

Om OS förändras, t.ex:

- Uppdatering
- Rootkit malware

Behöver det signeras på nytt

UEFI Secure Boot Problems…

Hur många olika OS finns det?

Signaturen är bara giltig om UEFI känner till den…

➔ Microsoft har näst intill monopol

eclypsium

THERE'S A HOLE IN THE BOOT

By: Eclypsium | July 29, 2020

"BootHole" vulnerability in the GRUB2 bootloader opens up Windows and Linux devices using Secure Boot to attack. All operating systems using GRUB2 with Secure Boot must release new installers and bootloaders.

https://eclypsium.com/blog/theres-a-hole-in-the-boot/

eclypsium

BOOTHOLE: HOW IT STARTED, HOW IT'S GOING

By: Eclypsium | April 14, 2021

It has been a little over nine months since Eclypsium researchers first published their findings on the **BootHole** vulnerability in the GRUB2 bootloader. BootHole garnered plenty of attention due to its potential to undermine the boot security of a device along with its incredibly wide reach, affecting virtually every Linux distribution as well as any device, including Windows devices, that used the industry-standard Microsoft Third Party UEFI Certificate Authority.

https://eclypsium.com/blog/boothole-how-it-started-how-its-going/

Ransomware

Dad Jokes
@Dadsaysjokes

"Hey officer, how did the hackers escape?"

No idea, they just ransomware.

10:47 PM · Jun 22, 2020 · Twitter for iPhone

---

57 Retweets **9** Quote Tweets **374** Likes

Screenshot of Wana Decrypt0r 2.0, by Anarchyte, 2017, Wikipedia (https://en.wikipedia.org/wiki/File:Wana_Decrypt0r_screenshot.png). Fair use/I enlighet med citaträtt, copyright most likely 라자루스 조직 (Lazarus Group, North Korea).

https://www.bbc.co.uk/news/health-39899646

The New York Times

Cyberattack Forces a Shutdown of a Top U.S. Pipeline

The operator, Colonial Pipeline, said it had halted systems for its 5,500 miles of pipeline after being hit by a ransomware attack.

Give this article

https://www.nytimes.com/2021/05/08/us/politics/cyberattack-colonial-pipeline.html

Cyber Security + Add to myFT

How Conti ransomware group crippled Costa Rica – then fell apart

Nation struggles in aftermath of president's refusal to pay to end cyber attack, even as hacking group collapsed

BLEEPINGCOMPUTER SIGN UP

Costa Rica's public health agency hit by Hive ransomware

By Sergiu Gatlan May 31, 2022 01:34 PM 0

KALIX KOMMUN

Kalix kommuns hantering av driftfel till följd av IT-attacken

Natten mot torsdag den 16 december gjorde någon intrång på Kalix kommuns servrar och krypterade data, vilket medförde allvarliga konsekvenser för hela verksamheten.

Konsekvenserna för våra verksamheter var allvarliga, men vi lyckades snabbt ställa om till analogt arbete. Det stod också snart klart att det handlade om en så kallad ransomware-attack, men i stället för att betala lösensumman släckte vi ned samtliga servrar och inledde ett omfattande återställningsarbete.

Förundersökningen har sedermera lagts ned, och när vi nu sammanfattar de totala kostnaderna kopplade till IT-attacken och det efterföljande arbetet med att höja säkerheten uppgår totalsumman till 2 265 300 kronor.

svt NYHETER

Lokalt

/ NORRBOTTEN

Miljonkostnader för Kalix kommun efter it-attacken Foto: Jimmy Bäckström/arkiv/SVT

Miljonkostnader för Kalix kommun efter it-attacken

UPPDATERAD 14 JANUARI 2022 PUBLICERAD 14 JANUARI 2022

Det har snart gått en månad sedan Kalix kommun drabbades av ett allvarligt driftfel på grund av en it-attack där hackarna krävde en lösensumma. Sedan dess har det pågått ett intensivt arbete med att åtgärda säkerheten.

– Det har aldrig varit aktuellt för oss att betala en lösensumma, då skulle vi vara villebråd igen, säger Maria Henriksson, kommundirektör på Kalix kommun.

Kalix kommun satsar nu på att bygga upp it-säkerheten och alla system ska nu vara åtgärdade men trots det fortsätter vissa saker att spöka för kommunen.

Researchers find new destructive wiper malware in Ukraine

It's been dubbed "CaddyWiper" by analysts at ESET

By Corin Faife | @corintxt | Mar 14, 2022, 5:23pm EDT

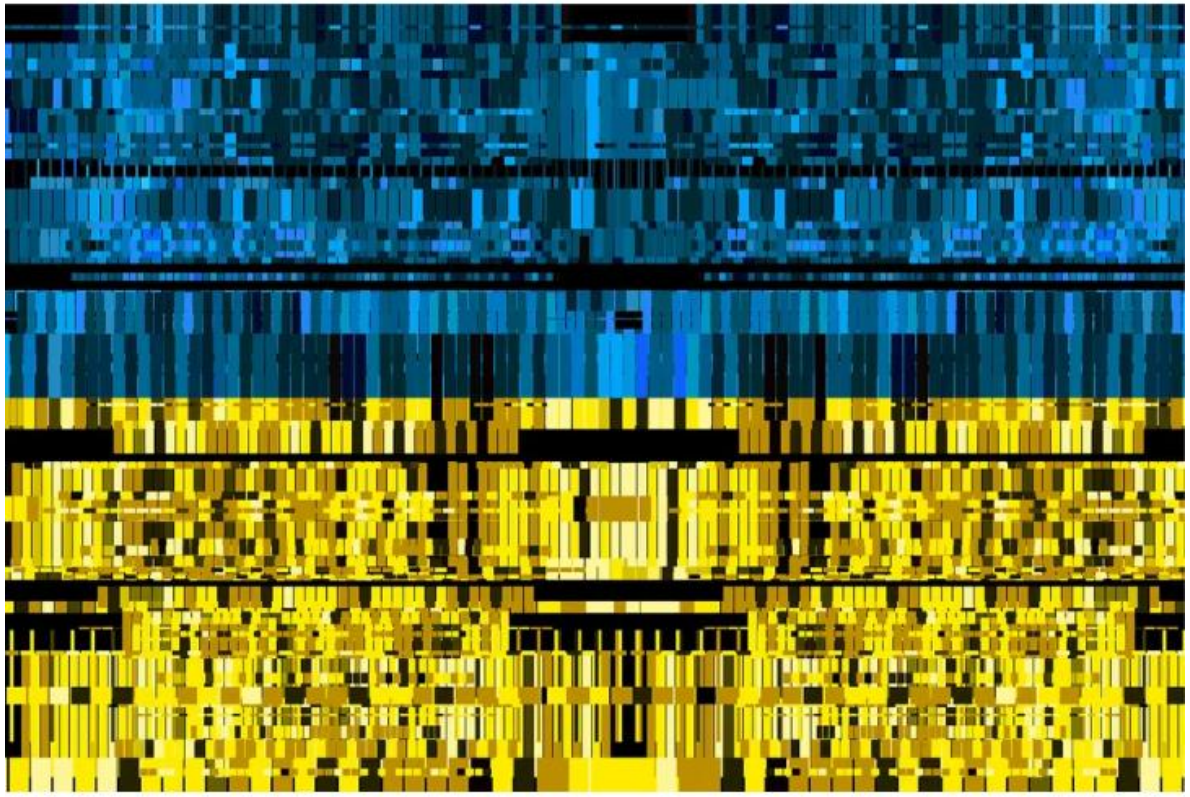


Illustration by Kristen Radtke / The Verge

https://www.theverge.com/2022/3/14/22977873/ukraine-new-destructive-caddywiper-malware-eset

Hur skyddar man sig mot ransomware?

Supply Chain Attacks

svt NYHETER

Nyheter | Lokalt | Sport | SVT Play | Barn

/ INRIKES

Se hur it-attacken drabbar landets Coop-butiker. Foto: Daniel Brodén/SVT

It-attack bakom Coops kassahaveri – stänger butikerna i hela Sverige

UPPDATERAD 16 JULI 2021 PUBLICERAD 2 JULI 2021

En it-attack mot en leverantör har fått Coops kassasystem att haverera. Coop rekommenderade samtliga butiker i hela Sverige att hålla stängt under lördagen – och även under söndagen hålls en majoritet av butikerna stängda.

Haveriet uppstod vid 19-tiden på fredagen och omfattar både självbetalningskassor och bemannade kassor. När SVT Nyheter pratade med Coop på lördagsmorgonen hade de tagit beslutet att hålla de allra flesta butikerna stängda.

Majoriteten av Coops nästan 800 butiker tvingas hålla stängt även under söndagen, säger Kevin Bell, pressekreterare för Coop.

https://www.svt.se/nyheter/inrikes/coop-tvingas-stanga-efter-kassahaveri

SvD Näringsliv

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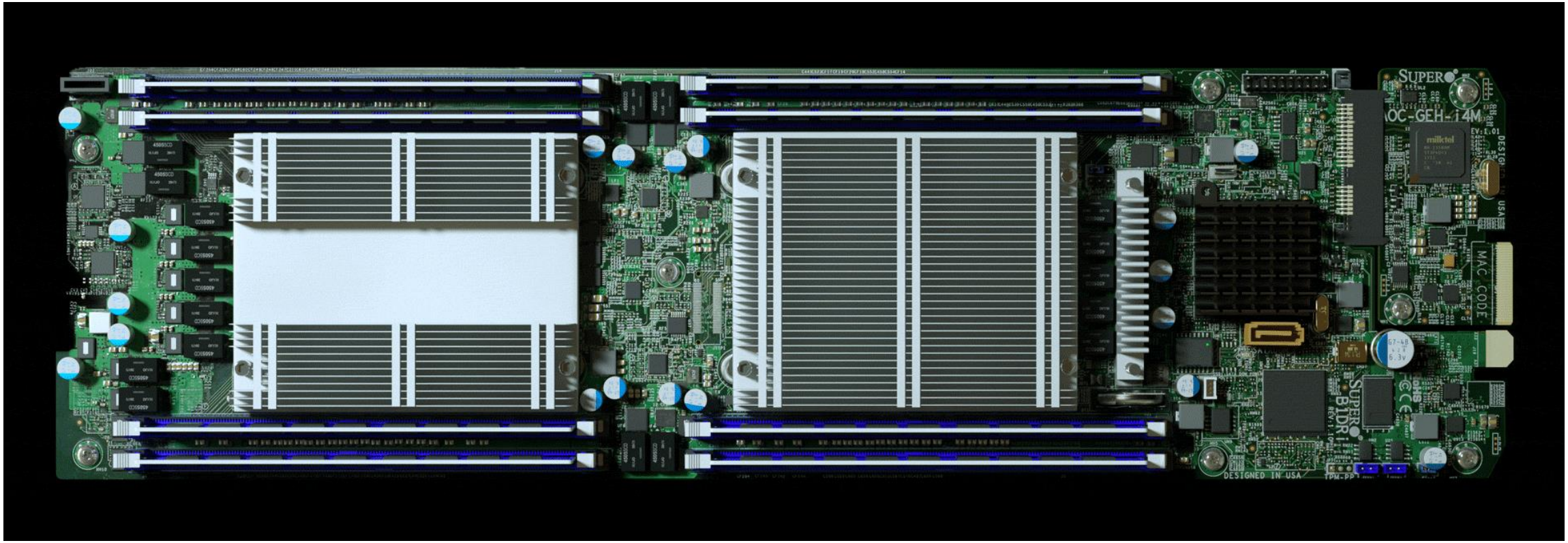
It-attacken mot Coop

Coopbutiker låsta i global it-attack

https://www.svd.se/a/1BQOOW/coop-i-hela-landet-stangs-efter-it-attack

Super Micro Computer logo, by Super Micro Computer, Wikimedia Commons (https://commons.wikimedia.org/wiki/File:Super_Micro_Computer_Logo.svg). Public domain, copyright of Super Micro Computer

https://www.bloomberg.com/news/features/2018-10-04/the-big-hack-how-china-used-a-tiny-chip-to-infiltrate-america-s-top-companies



https://www.bloomberg.com/news/features/2018-10-04/the-big-hack-how-china-used-a-tiny-chip-to-infiltrate-america-s-top-companies

Project Gunman

1976-1984

Amerikanska ambassaden i Soviet

https://www.cryptomuseum.com/covert/bugs/selectric/index.htm

NSA TAO: ANT CATALOG

TOP SECRET//COMINT//REL TO USA, FVEY

GODSURGE

ANT Product Data

(TS//SI//REL) GODSURGE runs on the FLUXBABBITT hardware implant and provides software application persistence on Dell PowerEdge servers by exploiting the JTAG debugging interface of the server's processors.

06/20/08

TOP SECRET//COMINT//REL TO USA, FVEY

IRATEMONK

ANT Product Data

(TS//SI//REL) IRATEMONK provides software application persistence on desktop and laptop computers by implanting the hard drive firmware to gain execution through Master Boot Record (MBR) substitution.

06/20/08

TOP SECRET//COMINT//REL TO USA, FVEY

SURLYSPAWN

ANT Product Data

(TS//SI//REL TO USA,FVEY) Data RF retro-reflector. Provides return modulated with target data (keyboard, low data rate digital device) when illuminated with radar.

07 Apr 2009

TOP SECRET//COMINT//REL TO USA, FVEY

RAGEMASTER

ANT Product Data

(TS//SI//REL TO USA,FVEY) RF retro-reflector that provides an enhanced radar cross-section for VAGRANT collection. It's concealed in a standard computer video graphics array (VGA) cable between the video card and video monitor. It's typically installed in the ferrite on the video cable.

24 Jul 2008

https://www.eff.org/files/2014/01/06/20131230-appelbaum-nsa_ant_catalog.pdf

Hur skyddar man sig mot planterade sårbarheter?

Bolagsverket
Konsumentverket · KO
Teknikföretagen
Sveriges Kommuner och Regioner
Statens servicecenter
IMY. Integritetsskyddsmyndigheten
PENSIONSMYNDIGHETEN
SSF Stöldskyddsföreningen
Telia
SVENSKT NÄRINGSLIV
företagarna
Verksamt.se
Till företag från myndigheter
Handelsbanken
TÄNK SÄKERT
Nordea
Konsumenternas
Kronofogden
Polisen
MSB Myndigheten för samhällsskydd och beredskap
ISACA Sweden Chapter
DATAFÖRENINGEN
Digital destination Sverige
Digitalidag
SäkerhetsBranschen
KUNSKAP · ANSVAR · KVALITET
SACS Swedish Association of Civil Security
Skatteverket
SH Svensk Handel
INTERNET STIFTELSEN

https://www.msb.se/sv/amnesomraden/informationssakerhet-cybersakerhet-och-sakra-kommunikationer/informationssakerhetsmanaden/

TÄNK SÄKERT

Meny

https://tanksakert.sakerhetskollen.se/

Till nästa gång

- Fredag: ingen laboration!
- Tentamen!
- Kursutvärdering

SOMEONE FIGURED OUT MY PASSWORD,
NOW I HAVE TO RENAME MY DOG.

Tack för idag